

# Notes

## Assignments

◆ 5.2, 5.5, 5.15, 5.21(a-f, h)

## Tutorial problems

● 5.1, 5.3, 5.4, 5.41

## The CT Fourier Transform Pair

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt \quad \text{— } FT$$

*Fourier Transform*

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t} d\omega \quad \text{— Inverse } FT$$

*Inverse Fourier Transform*

# With CTFT, now the frequency response of an LTI system makes complete sense

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longrightarrow \boxed{h(t)} \longrightarrow y(t) = \sum_{k=-\infty}^{\infty} H(jk\omega_0) a_k e^{jk\omega_0 t}$$

$$a_k \longrightarrow \underbrace{H(jk\omega_0) a_k}_{\text{"gain"}}$$

$$H(j\omega) = \int_{-\infty}^{+\infty} h(t) e^{-j\omega t} dt$$



Impulse response  $\xleftrightarrow{\mathcal{F}}$  Frequency response

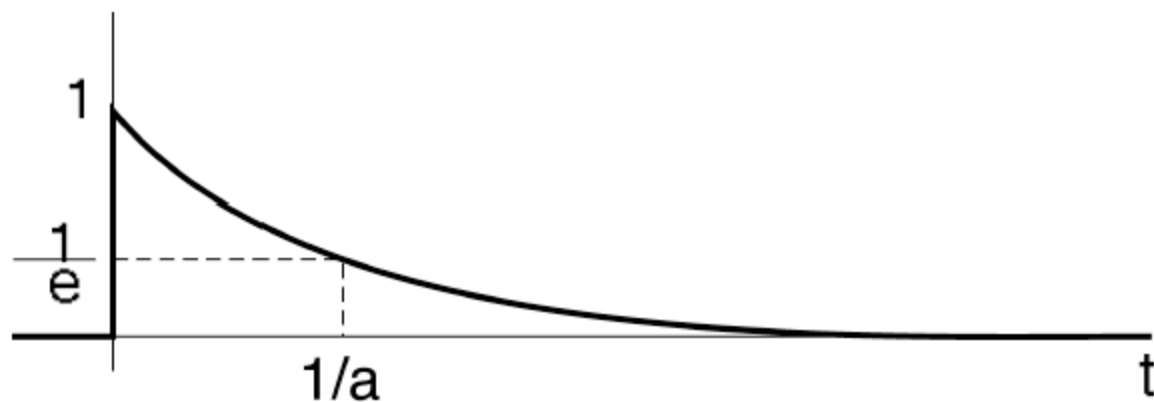
# CTFT Properties

## 8) Convolution Property

$$y(t) = h(t) * x(t) \longleftrightarrow Y(j\omega) = H(j\omega)X(j\omega)$$

$$\text{where } h(t) \longleftrightarrow H(j\omega) \quad x(t) \longleftrightarrow X(j\omega)$$

**Review** from the last lecture, right-sided exponential



$$x(t) = e^{-at}u(t) , \quad a > 0$$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt = \int_0^{+\infty} \underbrace{e^{-at}e^{-j\omega t}}_{e^{-(a+j\omega)t}} dt$$

$$= -\left(\frac{1}{a+j\omega}\right)e^{-(a+j\omega)t}\bigg|_0^{\infty} = \boxed{\frac{1}{a+j\omega}}$$

## Example 4.19

$$h(t) = e^{-t}u(t) \quad , \quad x(t) = e^{-2t}u(t)$$

$$y(t) = h(t) * x(t) = ?$$

$\Downarrow$

$$Y(j\omega) = H(j\omega)X(j\omega) = \frac{1}{(1+j\omega)} \cdot \frac{1}{(2+j\omega)}$$

$\Downarrow$  Partial fraction expansion

$$Y(j\omega) = \frac{1}{1+j\omega} \overset{a=1}{-} \frac{1}{2+j\omega} \overset{a=2}{+}$$

$\Downarrow$  inverse FT

$$y(t) = [e^{-t} - e^{-2t}]u(t)$$

## CTFT Properties

### 9) Multiplication Property

Since *FT* is highly symmetric,

$$x(t) = \mathcal{F}^{-1}\{X(j\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega, \quad X(j\omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

thus if

$$x(t) * y(t) \longleftrightarrow X(j\omega) \cdot Y(j\omega)$$

then the other way  
around is also true

$$\begin{aligned} x(t) \cdot y(t) &\longleftrightarrow \frac{1}{2\pi} X(j\omega) * Y(j\omega) \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta) Y(j(\omega - \theta)) d\theta \end{aligned}$$

$\frac{1}{2\pi}$

— A consequence of *Duality*

# Examples of the Multiplication Property: Modulation Property

Frequency shift

$$e^{j\omega_0 t} x(t) \xleftrightarrow{F} X(j(\omega - \omega_0))$$

$$e^{j\omega_0 t} \cdot x(t) \longleftrightarrow \frac{1}{2\pi} [2\pi\delta(\omega - \omega_0) * X(j\omega)] \\ = X(j(\omega - \omega_0))$$

## Example 4.21

$$r(t) = s(t) \cdot p(t) \longleftrightarrow R(j\omega) = \frac{1}{2\pi} [S(j\omega) * P(j\omega)]$$

$$\text{For } p(t) = \cos\omega_0 t \longleftrightarrow P(j\omega) = \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$

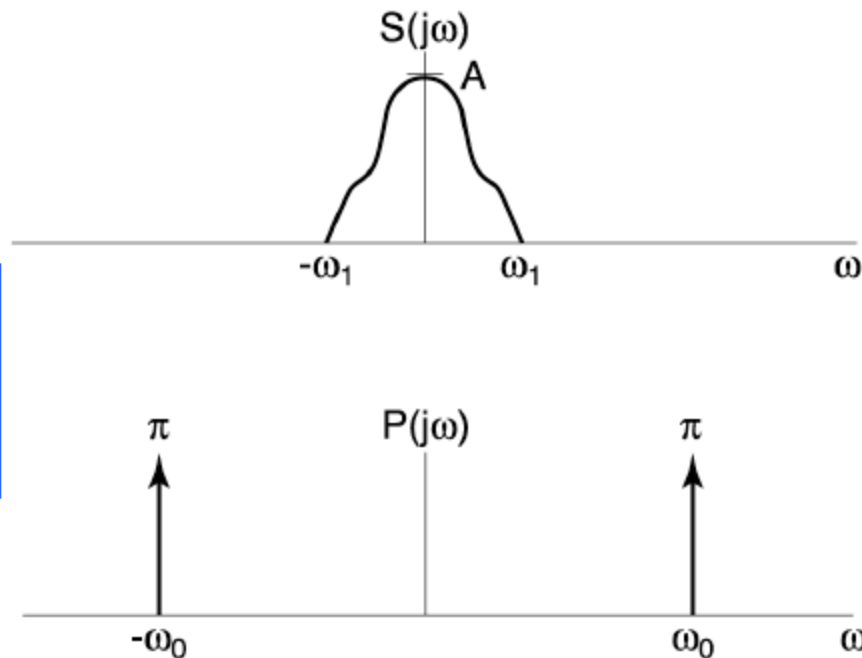
$$R(j\omega) = \frac{1}{2} S(j(\omega - \omega_0)) + \frac{1}{2} S(j(\omega + \omega_0))$$



# (cont.)

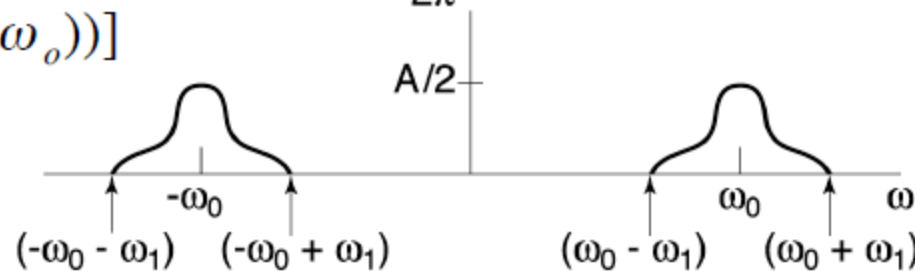
$\omega_1$  : bandwidth

$r(t) = s(t) \cdot \cos(\omega_o t)$   
Amplitude  
modulation (*AM*)



$$R(j\omega) = \frac{1}{2} [S(j(\omega - \omega_o)) + S(j(\omega + \omega_o))]$$

$$R(j\omega) = \frac{1}{2\pi} [S(j\omega) * P(j\omega)]$$



Drawn assume  
 $\omega_o - \omega_1 > 0$   
i.e.  $\omega_o > \omega_1$

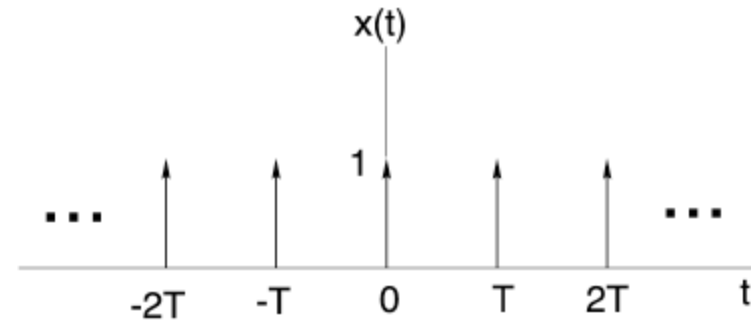
# Example 4.8

$$x(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT) \quad \text{— sampling function}$$

$$x(t) \xleftrightarrow{\text{FS}} a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_o t} dt = \frac{1}{T}$$

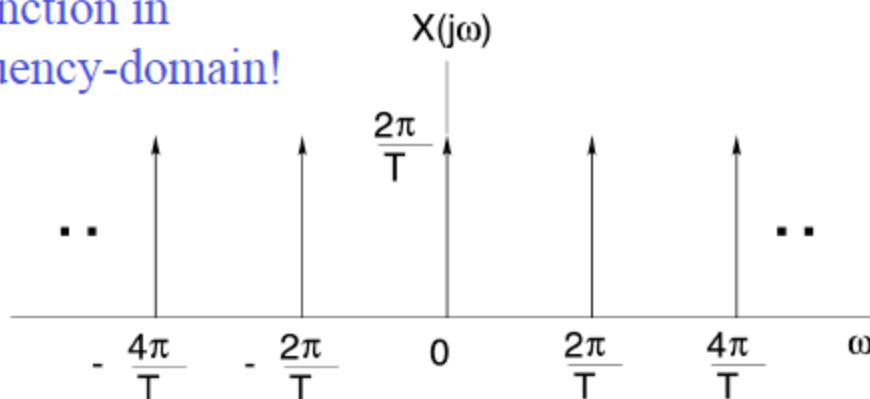
$$\Downarrow x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_o t} = \sum_{k=-\infty}^{+\infty} \frac{1}{T} e^{jk\omega_o t}$$

$$X(j\omega) = \sum_{k=-\infty}^{+\infty} \frac{2\pi}{T} \delta(\omega - \underbrace{\frac{k2\pi}{T}}_{k\omega_o})$$



$\omega_s = 2\pi / T$  : sampling frequency

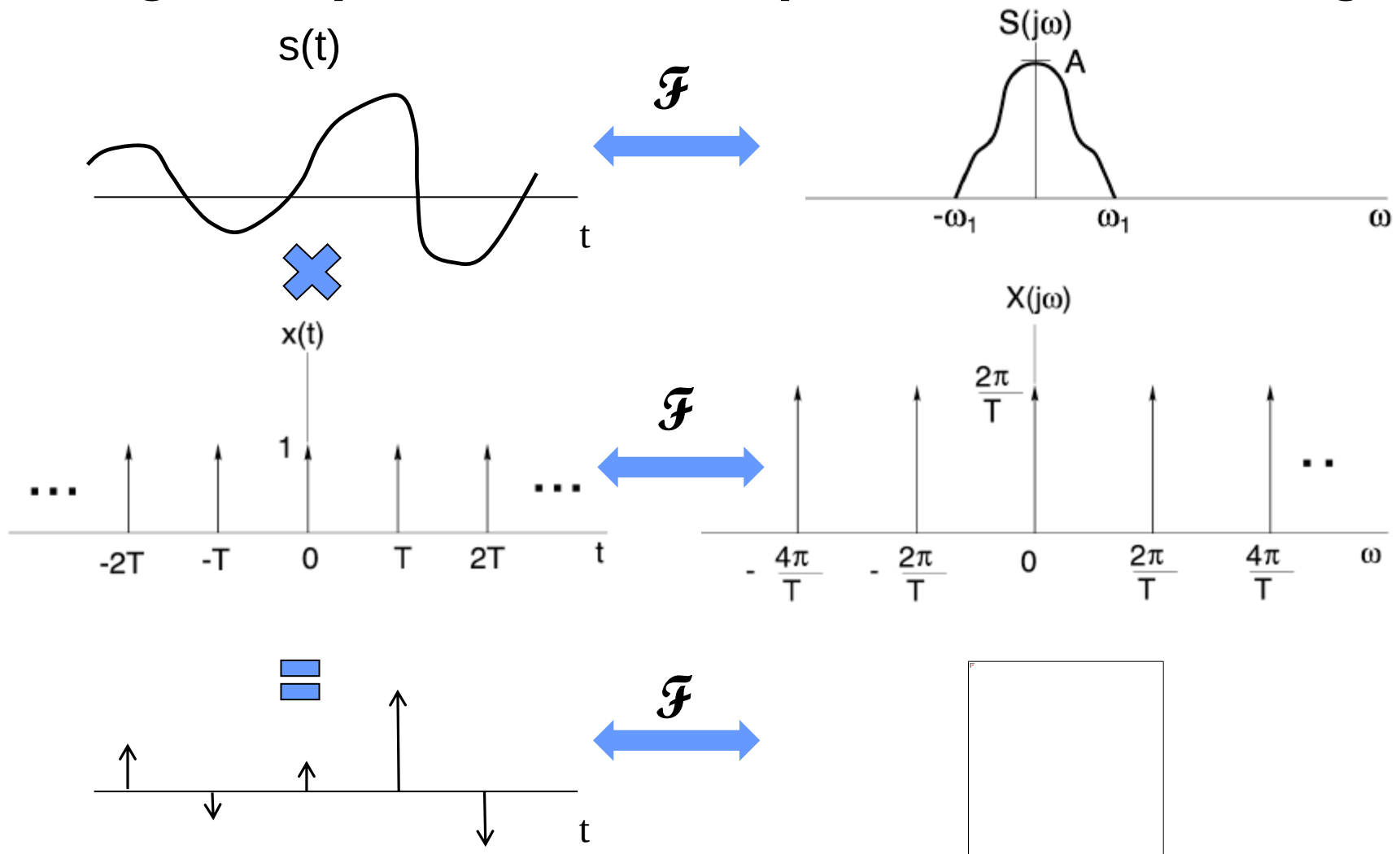
Same function in the frequency-domain!



Note in this case, periodic in both time domain (with a period  $T$ ) and frequency domain (with a period  $2\pi/T$ )

(cont.)

- Image the process to sample a continuous signal



## Example #8: LTI Systems Described by LCCDE's (Linear-constant-coefficient differential equations)

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

Using the Differentiation Property

$$\frac{d^k x(t)}{dt^k} \longleftrightarrow (j\omega)^k X(j\omega)$$

$\Downarrow$  Transform both sides of the equation

$$\sum_{k=0}^N a_k (j\omega)^k Y(j\omega) = \sum_{k=0}^M b_k (j\omega)^k X(j\omega)$$

$\Downarrow$

$$Y(j\omega) = \underbrace{\left[ \frac{\sum_{k=0}^M b_k (j\omega)^k}{\sum_{k=0}^N a_k (j\omega)^k} \right]}_{H(j\omega)} X(j\omega)$$

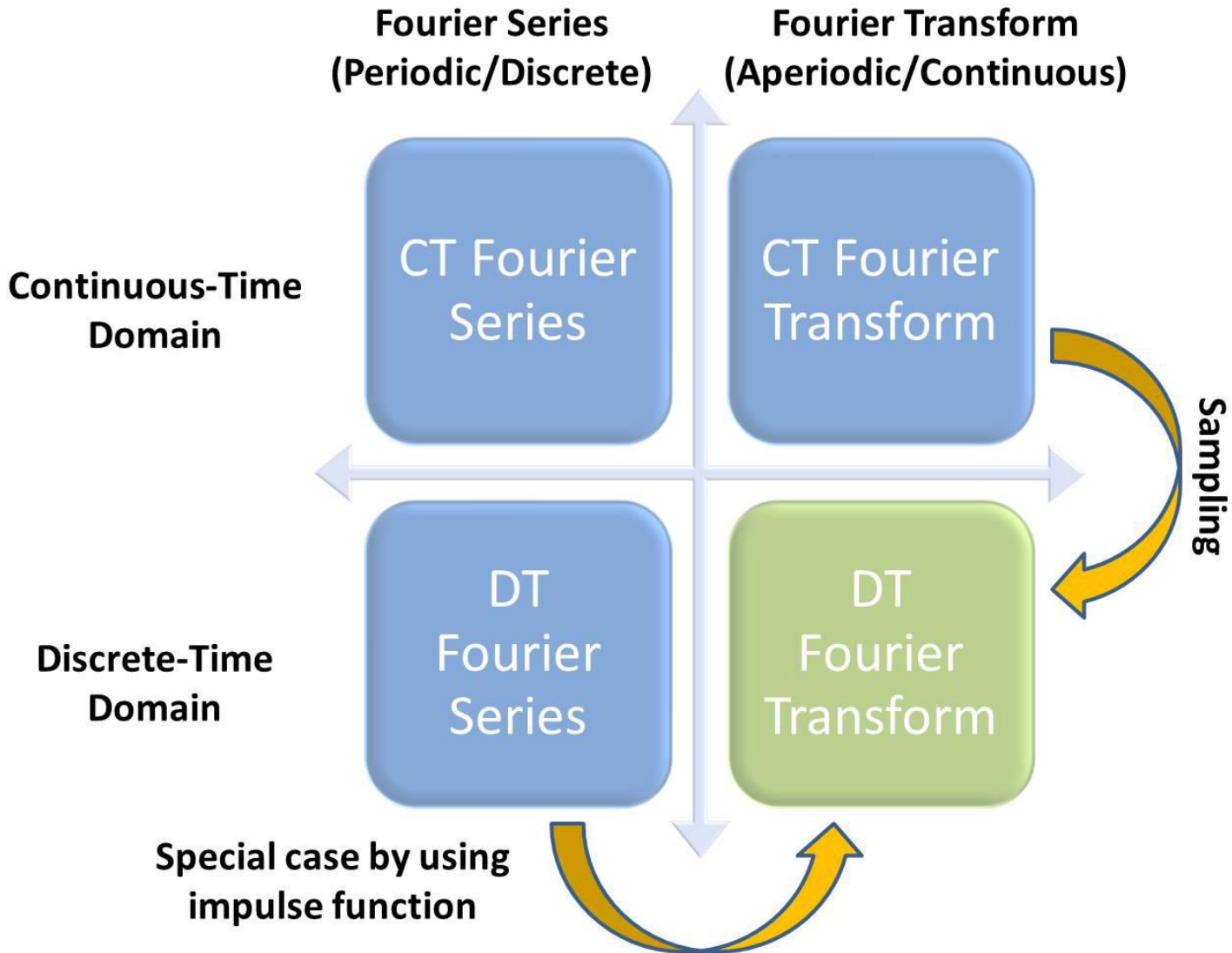
$$H(j\omega) = \left[ \frac{\sum_{k=0}^M b_k (j\omega)^k}{\sum_{k=0}^N a_k (j\omega)^k} \right]$$

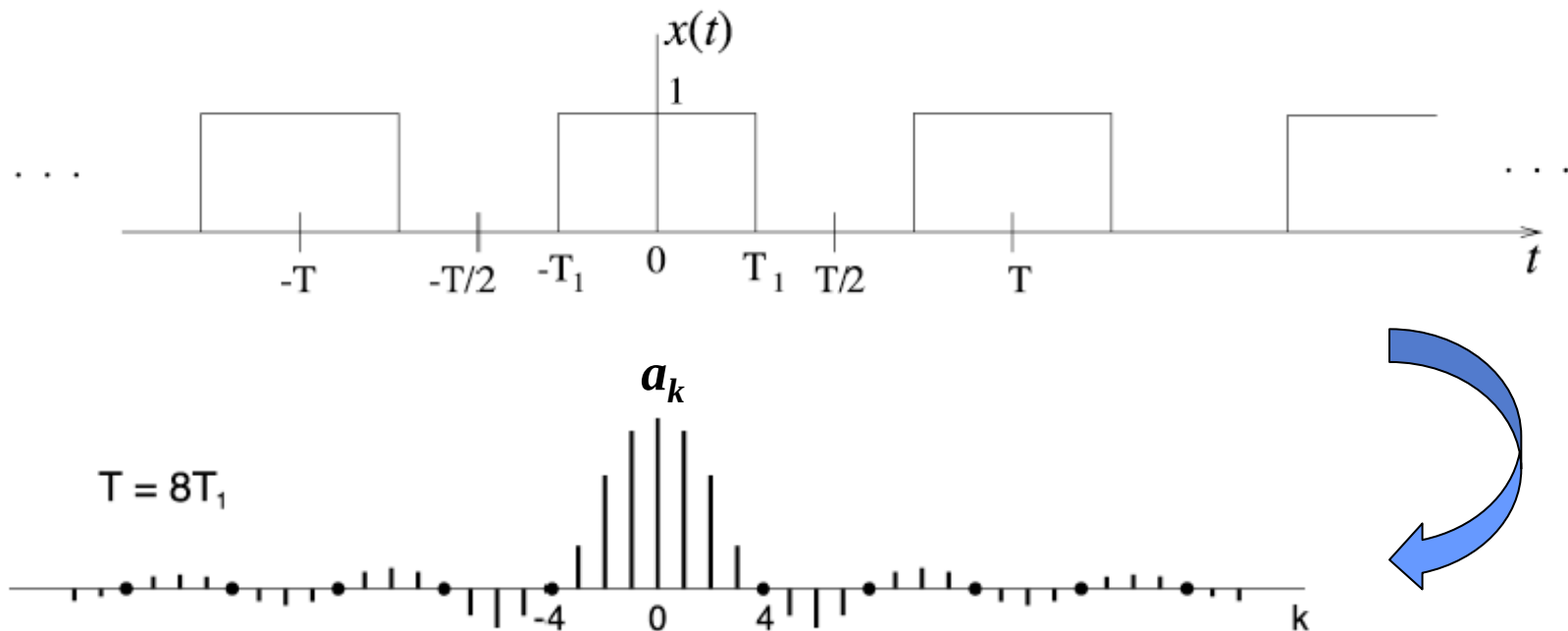
# **Chapter 5**

## **The Discrete-Time Fourier Transform**

# Frequency domain

Time domain





- The harmonic components are spaced  $\omega_0 = 2\pi/T$  apart, as  $T \rightarrow \infty$ , and  $\omega_0 \rightarrow 0$ , then  $\omega = k\omega_0$  becomes continuous



Fourier series  $\longrightarrow$  Fourier integral

# Thinking...

- How to move from **CT Fourier series** to **CT Fourier transform**?



# Discrete-Time Fourier Transform (DTFT)

## DT Fourier Series Pair $\left(\omega_o = \frac{2\pi}{N}\right)$

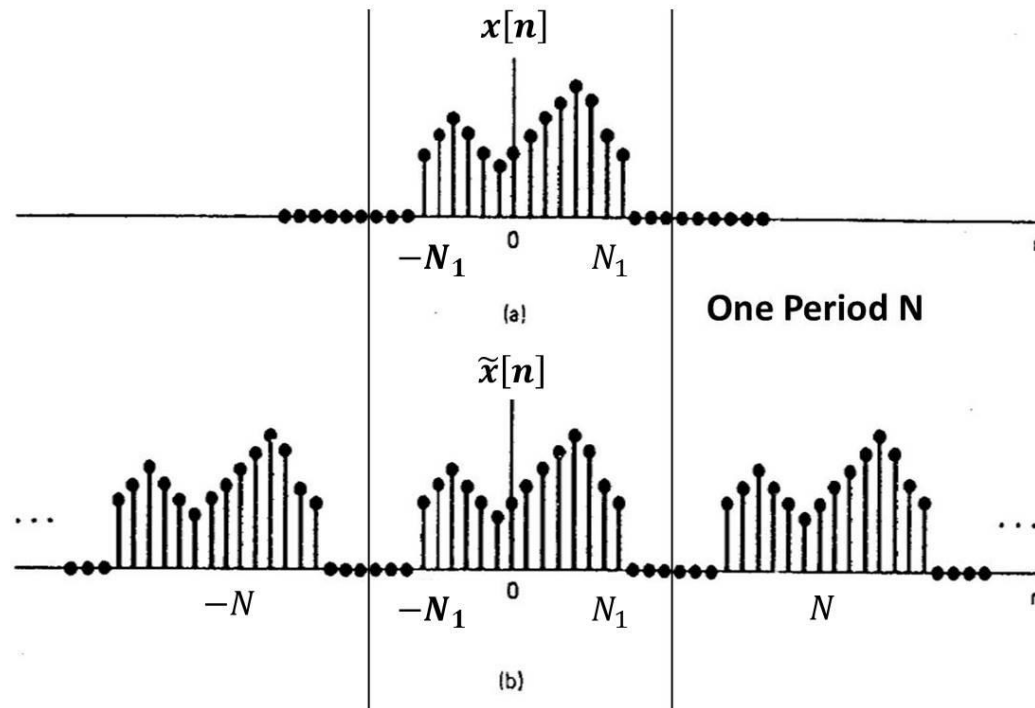
$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_o n} \quad (\text{Synthesis equation})$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_o n} \quad (\text{Analysis equation})$$

Aperiodic signals can be treated as periodic signals with period  $N \rightarrow \infty$

- ▶  $x[n]$  must be like  $\sum_k b_k e^{jk(2\pi/N)n}$  or  $\int_{\omega} b(\omega) e^{j\omega n} d\omega$
- ▶  $b_k$  or  $b(\omega)$  can be calculated from  $x[n]$

# DTFT Derivation (1/3)



Original signal:  $x[n]$

Define new periodic signal with period  $N$ :  $\tilde{x}[n]$ , such that

$$\tilde{x}[n] = x[n], \quad n = -N/2, \dots, N/2 - 1$$

Notice: when  $N \rightarrow \infty$ ,  $\tilde{x}[n]$  becomes  $x[n]$

## DTFT Derivation (2/3)

- Look at the Fourier series of  $\tilde{x}[n]$ :

$$a_k = \frac{1}{N} \sum_{n=-N/2}^{N/2+1} \tilde{x}[n] e^{-jk(2\pi/N)n} = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk(2\pi/N)n}$$

## DTFT Derivation (3/3)

- Therefore, we get the discrete-time Fourier transform pair

### Discrete-Time Fourier Transform

$$\text{Synthesis Equation: } x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$\text{Analysis Equation: } X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

# Periodicity Properties of DT Complex Exponentials

- $x[n]$  - periodic with fundamental period  $N$ , fundamental frequency

$$x[n + N] = x[n] \quad \text{and} \quad \omega_o = \frac{2\pi}{N}$$

$$n = \dots, -1, 0, 1, 2, 3, \dots$$

- For DT complex exponentials, signal are periodic only when  $\omega_0 N = k \cdot 2\pi$ ,  $k = 0, \pm 1, \pm 2, \dots$

$$e^{j\omega_0 n} = e^{j\omega_0 (n+N)} \rightarrow e^{j\omega_0 N} = 1 \rightarrow \omega_0 N = k \cdot 2\pi$$

- For DT complex exponentials, signals with frequencies  $\omega_0$  and  $\omega_0 + k \cdot 2\pi$  are identical.  $e^{j(\omega_0 + k \cdot 2\pi)n} = e^{j\omega_0 n} \cdot e^{jk \cdot 2\pi n} = e^{j\omega_0 n}$ 
  - We need only consider a frequency interval of length  $2\pi$ , and on most cases, we use the interval:  $0 \leq \omega_0 < 2\pi$ , or  $-\pi \leq \omega_0 < \pi$

## Cont.

-  $e^{j\omega_0 n}$  does **not** have a continually increasing rate of oscillation as  $\omega_0$  is increased in magnitude.

**low-frequency** (slowly varying):  $\omega_0$  near  $0, 2\pi, \dots$ , or  $2k \cdot \pi$

**high-frequency** (rapid variation):  $\omega_0$  near  $\pm \pi, \dots$ , or  $(2k+1) \cdot \pi$

$$e^{j(2k+1)\pi n} = e^{j\pi n} = (e^{j\pi})^n = (-1)^n$$

## Discrete-Time Fourier Transform

$$\text{Synthesis Equation: } x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

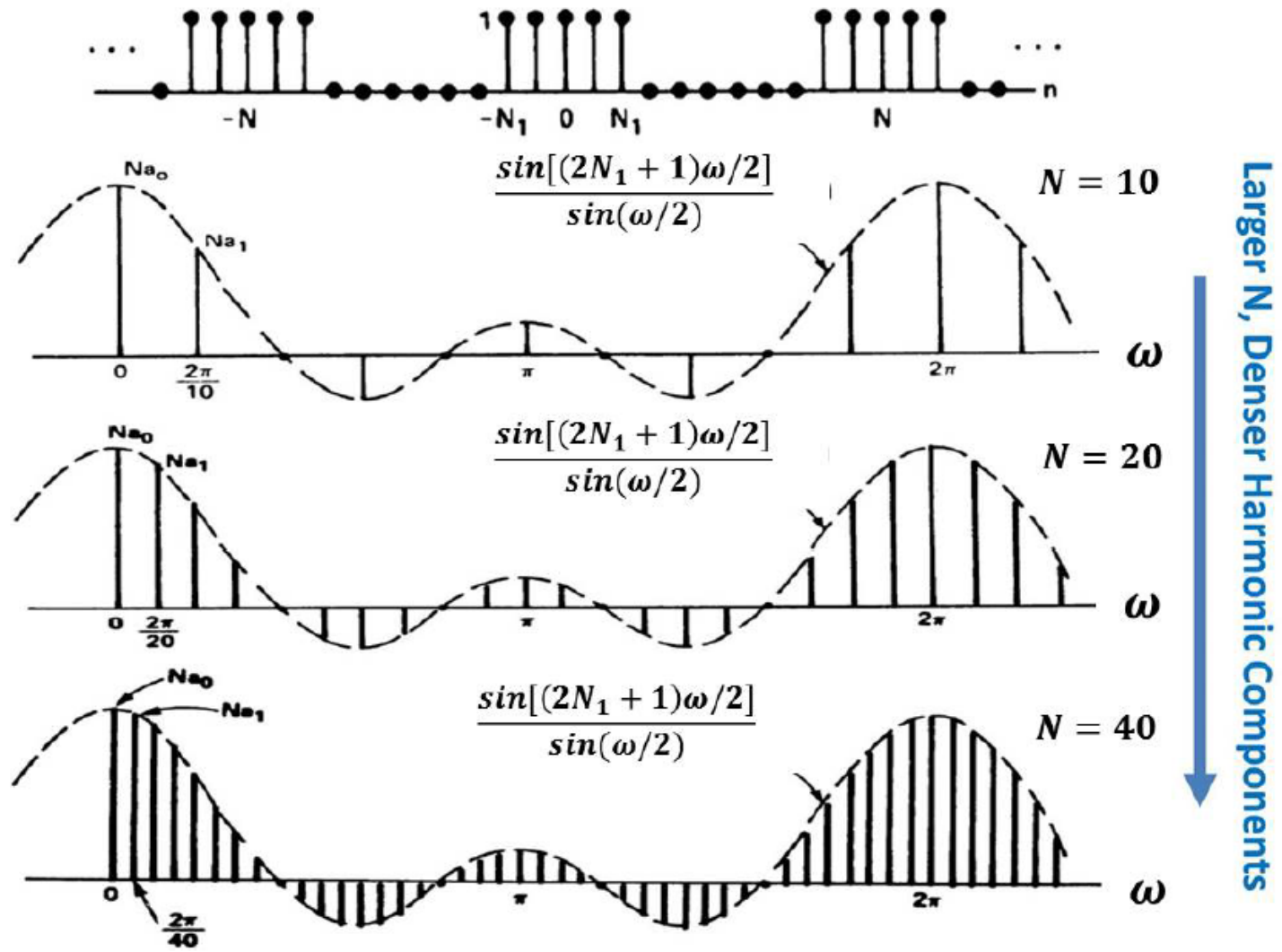
$$\text{Analysis Equation: } X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

## DT Fourier Series Pair $\left(\omega_o = \frac{2\pi}{N}\right)$

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_o n} \quad (\text{Synthesis equation})$$

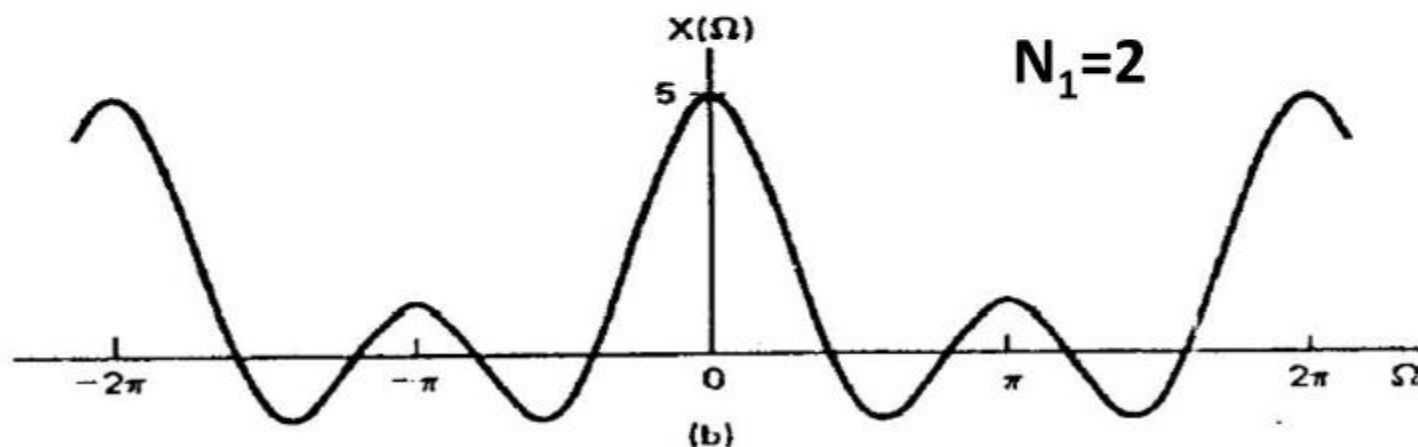
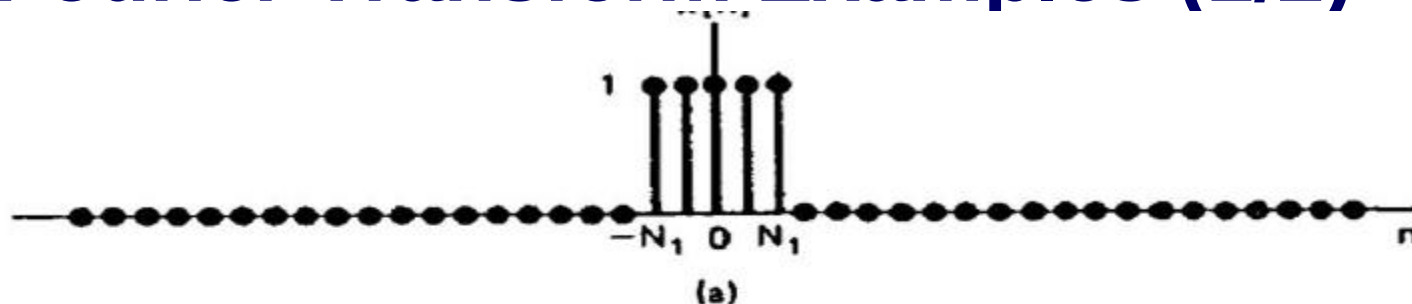
$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_o n} \quad (\text{Analysis equation})$$

# Example: From Periodic To Aperiodic





# Fourier Transform Examples (1/2)

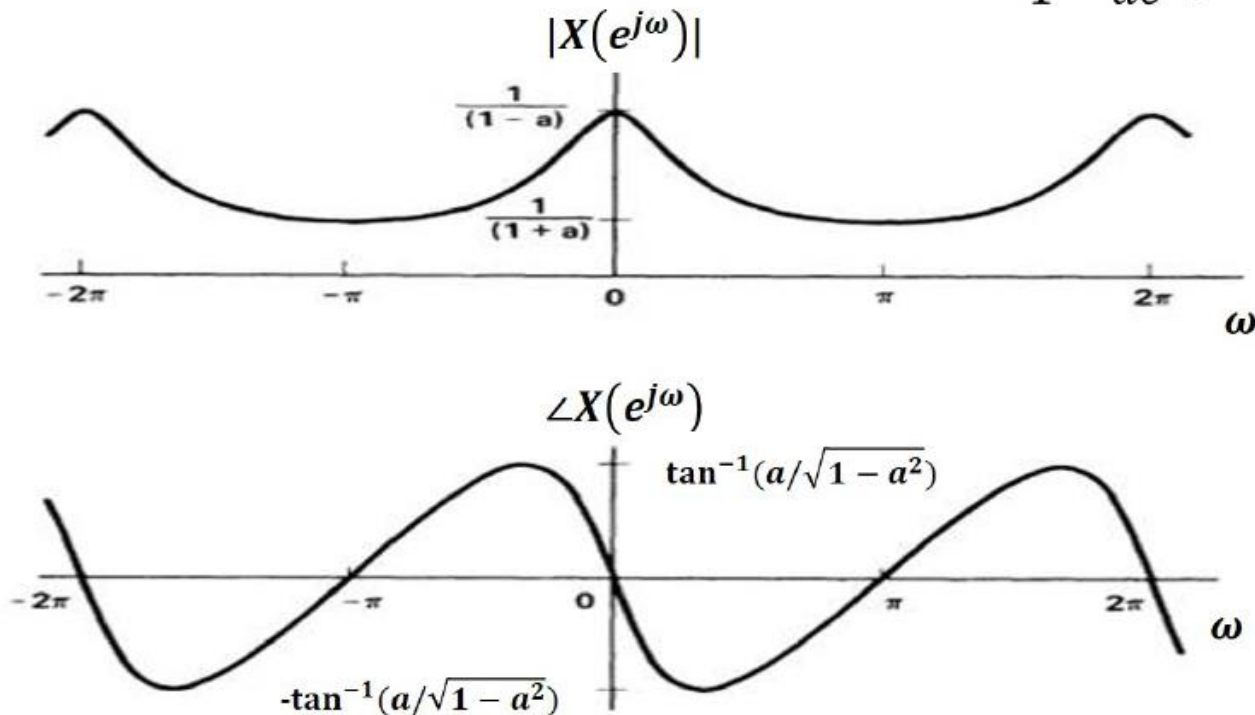


- $x[n] = 1$  ( $n = -N_1, \dots, 0, \dots, N_1$ )
- $X(e^{j\omega}) = \frac{\sin \omega(N_1+1/2)}{\sin(\omega/2)}$
- Width of  $x[n]$ :  $W_t = 2N_1 + 1$ ; width of  $X(e^{j\omega})$ :  $W_f = \frac{4\pi}{2N_1+1}$
- $W_t \times W_f = 4\pi$ , which is a constant



## Fourier Transform Examples (2/2)

$$x[n] = a^n u[n] \quad 0 < a < 1 \quad \Leftrightarrow \quad X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$$



- See textbook, Example 5.1
- What's the shape of magnitude when  $a \rightarrow 1$  or  $a \rightarrow 0$ ?

# Convergence Issue of Analysis Equation

## Sufficient Condition of Convergence

*The analysis equation  $X(e^{j\omega}) = \sum_{-\infty}^{\infty} x[n]e^{-j\omega n}$  will converge either if  $x[n]$  is absolutely summable or if the sequence has finite energy, thus,*

$$\sum_{-\infty}^{\infty} |x[n]| < \infty \text{ or } \sum_{-\infty}^{\infty} |x[n]|^2 < \infty$$

## Cont.

- Do the following signals have Fourier transform:
  - ▶  $a^n u[n]$  ( $0 < a < 1$ )
  - ▶  $\delta[n]$
  - ▶  $u[n]$
  - ▶  $e^{j\frac{2}{5}\pi n}$ ,  $\cos(\frac{2}{5}\pi n)$
  - ▶  $a^n u[n]$  ( $a > 1$ )

# Can Periodic Signals Have DTFT?

- Definition of DTFT:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

- Justification of divergence:

- ▶ Let  $\omega = 2k\pi$ , we have  $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]$

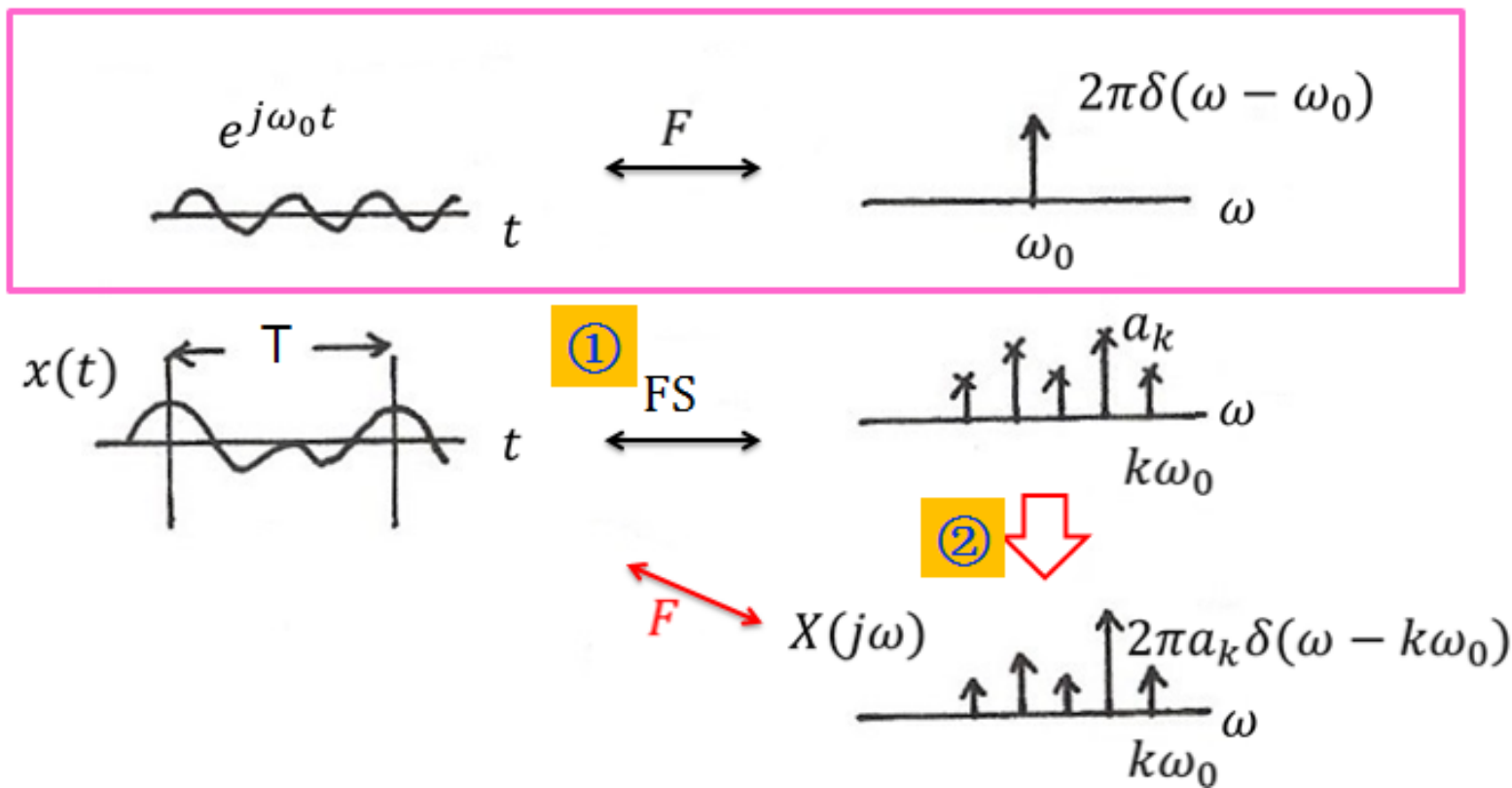
- ▶ Since  $x[n]$  is periodic, the summation  $\sum_{n=-\infty}^{\infty} x[n]$  will never converge unless  $x[n] = 0$

- **Conclusion:** Most of periodic signals do NOT have DTFT according to the definition
- However, it's of significant engineering importance to extend Fourier transform to periodic signals

# Can Periodic Signals Have DTFT?

## Review

### Fourier Transform for **Periodic Signals** – Unified Framework



# DTFT with Periodic Signals (1/2)

## Fourier Transform of $e^{j\omega n}$

*The following transform pair is actually NOT rigorously defined:*

$$x[n] = e^{j\omega_0 n} \longleftrightarrow X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 - 2\pi l)$$

- Synthesis:

$$\frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{2\pi} \sum_{l=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 - 2\pi l) e^{j\omega n} d\omega = e^{j\omega_0 n}$$

- Analysis:

$$\sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} e^{j(\omega_0 - \omega)n} \quad \text{converge??}$$

## DTFT with Periodic Signals (2/2)

- According to the Fourier series, for a periodic signal with period  $N$ :

$$x[n] = a_0 + a_1 e^{j(2\pi/N)n} + \dots + a_k e^{jk(2\pi/N)n} + \dots + a_{N-1} e^{j(N-1)(2\pi/N)n}$$

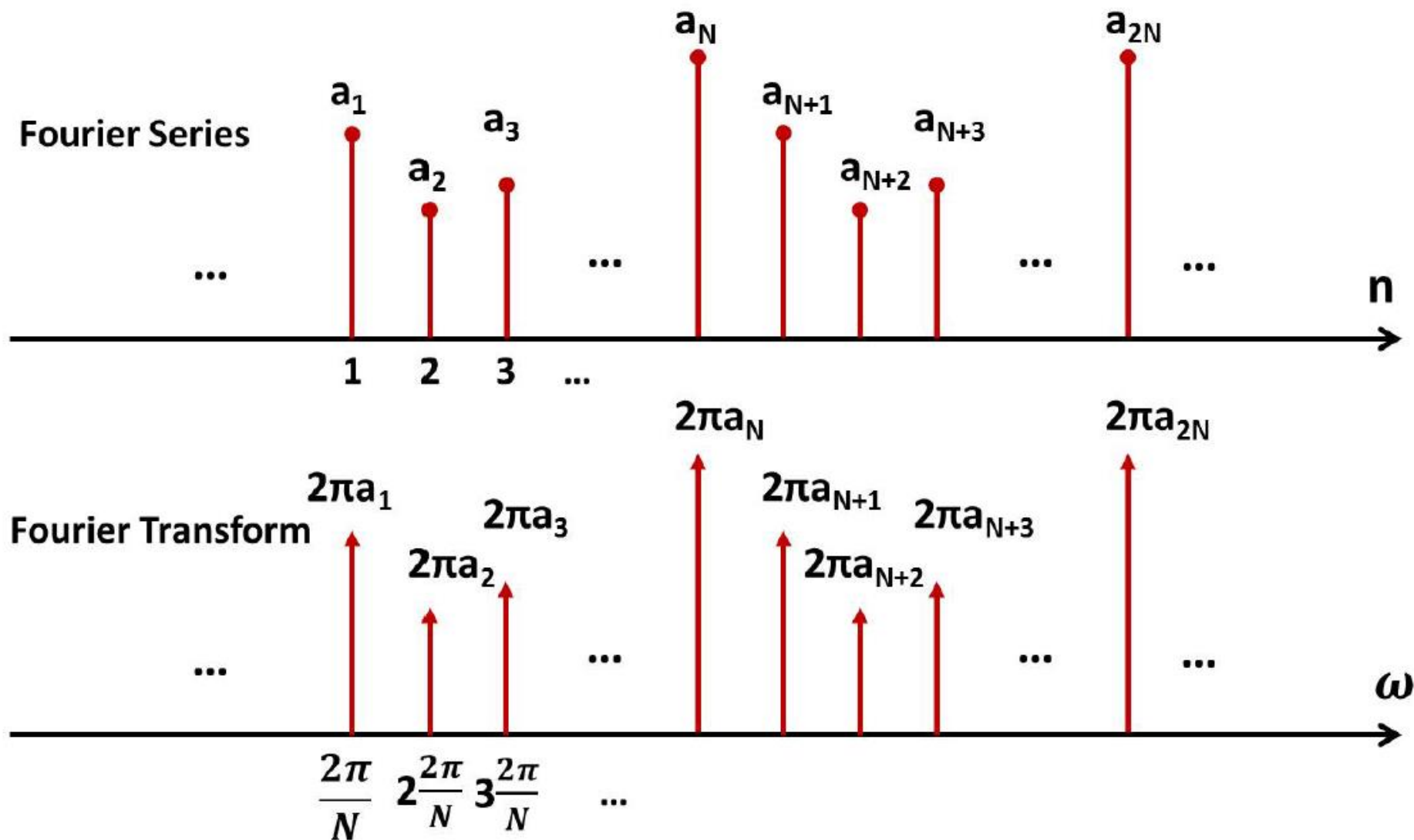
- $e^{jk(2\pi/N)n} \longleftrightarrow \sum_{l=-\infty}^{\infty} 2\pi \delta(\omega - k(2\pi/N) - 2\pi l)$
- Then, due to the linearity of Fourier transform

$$\begin{aligned} \mathcal{F}\{x[n]\} &= \sum_{k=0}^{N-1} a_k \mathcal{F}\{e^{jk(2\pi/N)n}\} = \sum_{k=0}^{N-1} a_k \sum_{l=-\infty}^{\infty} 2\pi \delta(\omega - k(2\pi/N) - 2\pi l) \\ &= \sum_{l=-\infty}^{\infty} \sum_{k=0}^{N-1} a_k 2\pi \delta(\omega - k(2\pi/N) - 2\pi l) \end{aligned}$$

- Fourier transform of a periodic signal is a periodic sequence of impulses
  - What's the period? How many impulses within one period?



# Fourier Series v.s. Fourier Transform



# Example: Discrete-Time Impulse Chain

- What's the Fourier transform of  $x[n] = \sum_{k=-\infty}^{+\infty} \delta[n - kN]$  ?
- First of all, we calculate the Fourier series:

$$\begin{aligned}
 a_k &= \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk \frac{2\pi}{N} n} \\
 &= \frac{1}{N} \sum_{n=\langle N \rangle} \sum_{k=-\infty}^{+\infty} \delta[n - kN] e^{-jk \frac{2\pi}{N} n} \\
 &= \frac{1}{N} \sum_{n=\langle N \rangle} \delta[n] e^{-jk \frac{2\pi}{N} n} = \frac{1}{N}
 \end{aligned}$$

# Frequency domain

Time domain

